

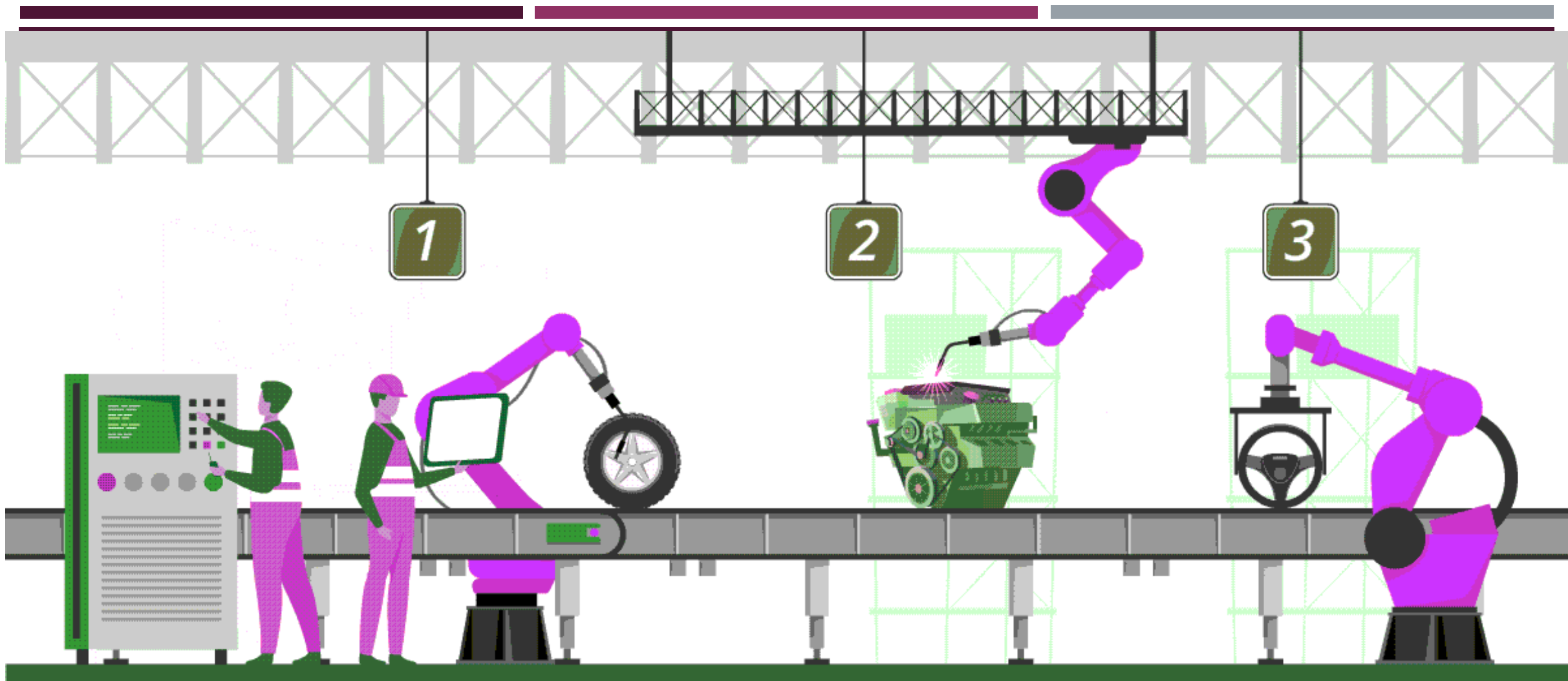


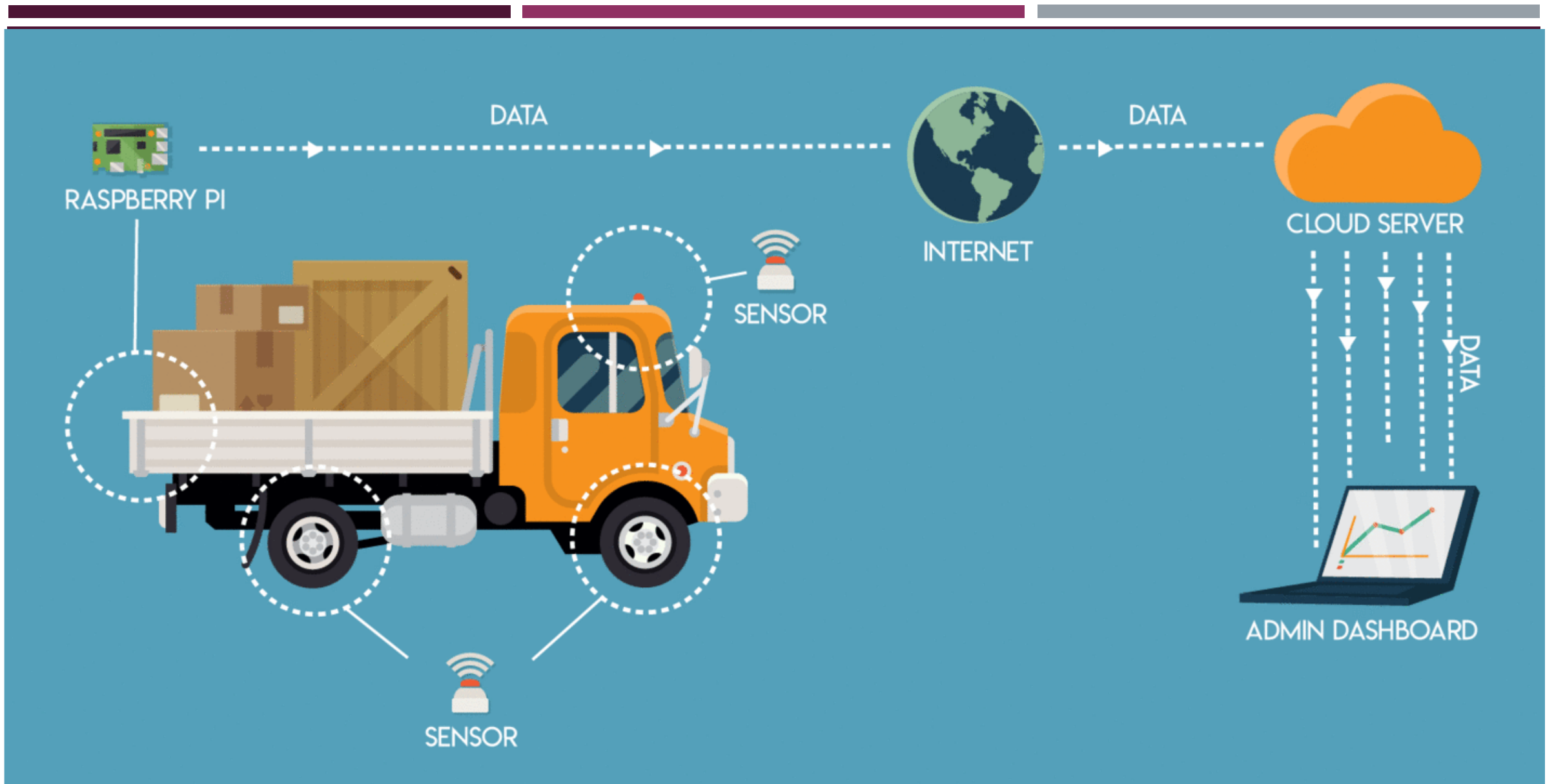
INTERNET OF THINGS

DR. AMANDEEP KAUR

DEPARTMENT OF MANAGEMENT STUDIES







INTERNET OF THINGS

- Internet of Things means a **network of physical things (objects)** sending, receiving, or communicating information using the Internet or other communication technologies and network just as the computers, tablets and mobiles do, and thus **enabling the monitoring, coordinating or controlling** process across the Internet or another data network.
- Internet of Things is the network of *physical objects or 'things'* embedded with electronics, software, sensors and connectivity to enable it to achieve greater value and service by exchanging data with the manufacturer, operator and/or other connected devices. Each thing is uniquely identifiable through its embedded computing system but is able to interoperate within the existing Internet infrastructure.

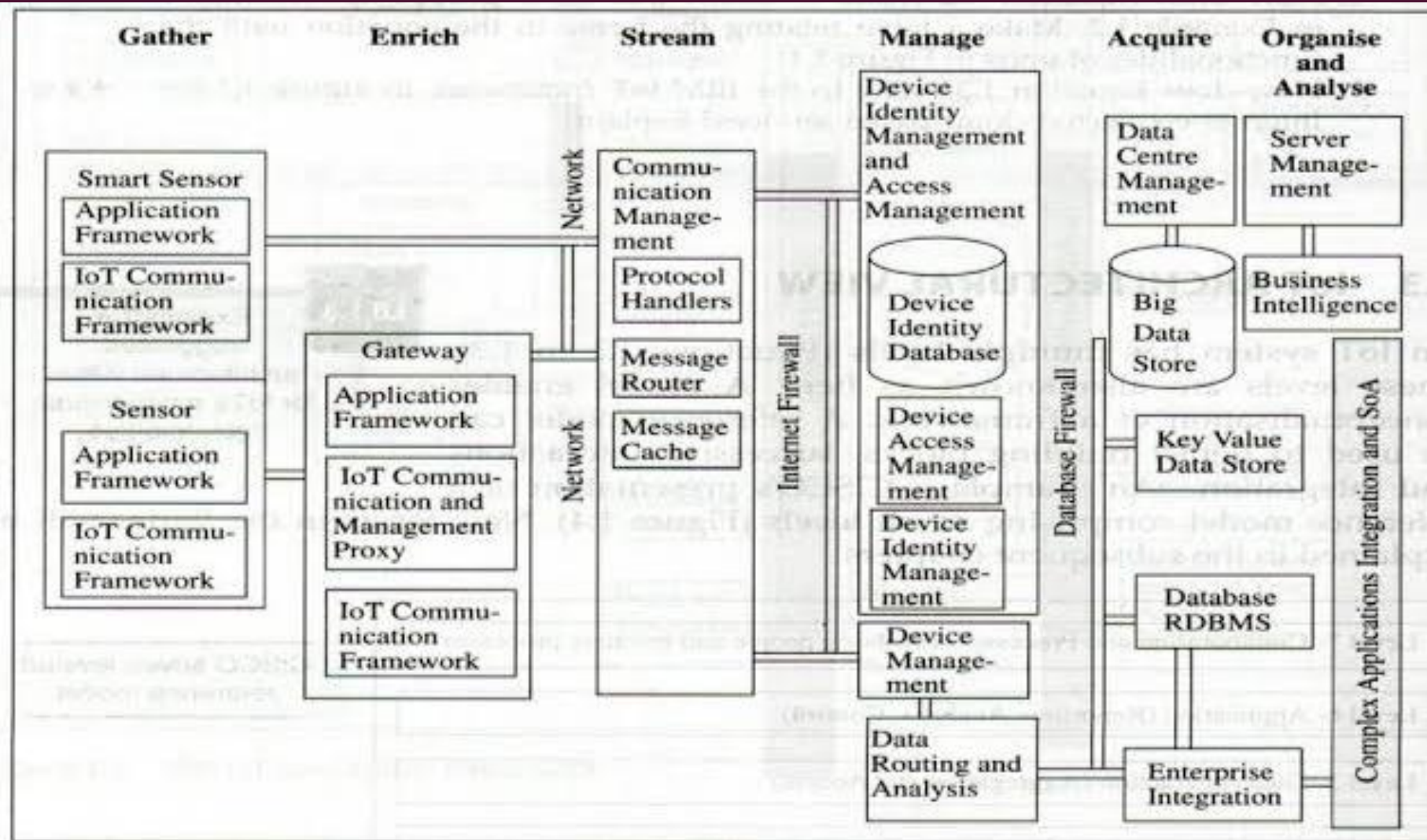
IOT CONCEPTUAL FRAMEWORK

- **Physical Object + Controller, Sensor and Actuators + Internet = Internet of Things**

(As per Oracle suggested IoT architecture)

- **Gather + Enrich + Stream + Manage + Acquire + Organise and Analyse = Internet of Things with connectivity to data centre, enterprise or cloud servers**

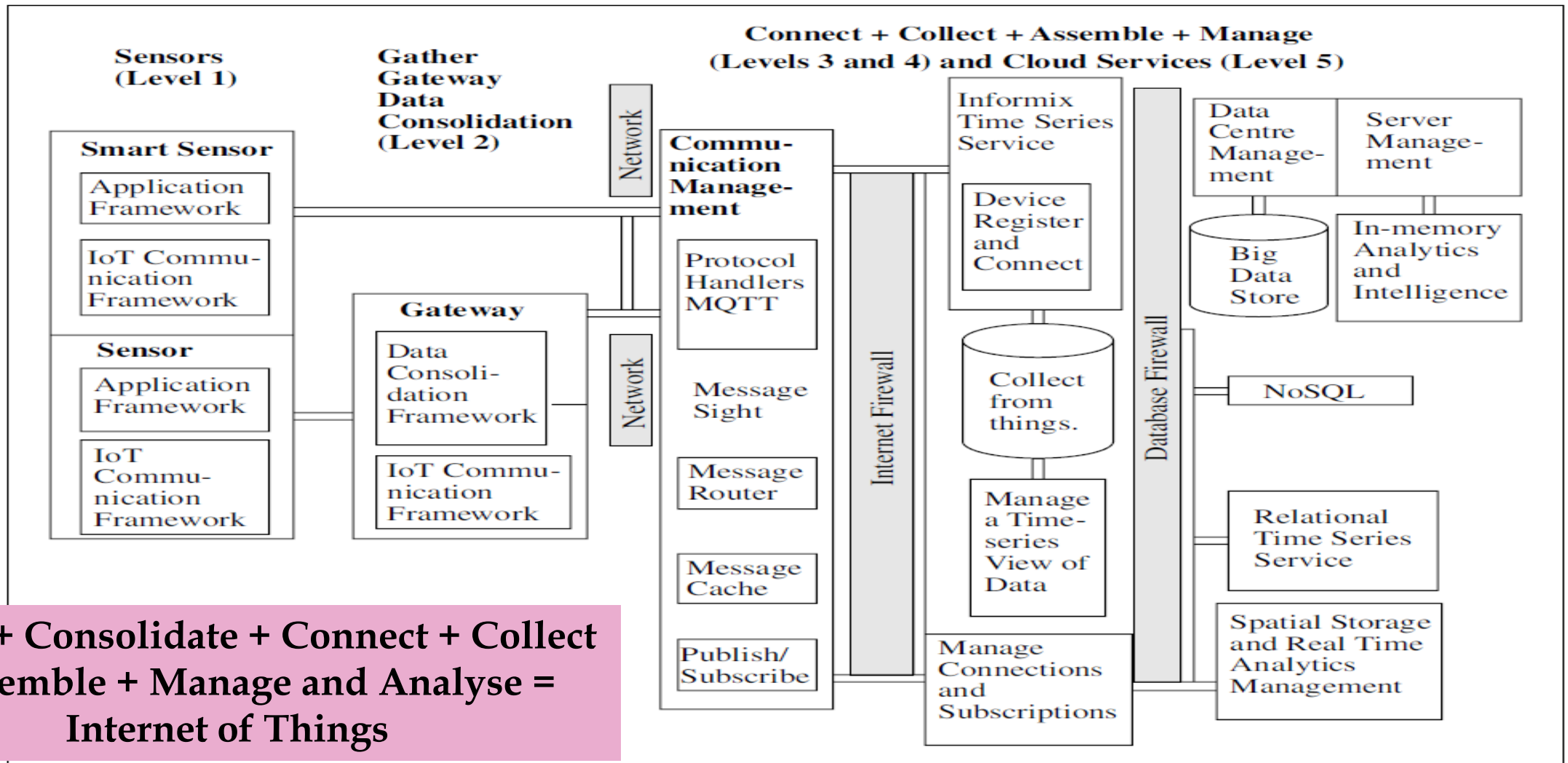
ORACLE IOT ARCHITECTURE



ORACLE IOT ARCHITECTURE

- **Gather:** Collects data from sensors and smart devices.
- **Enrich:** Processes and converts sensor data through the gateway.
- **Stream:** Transmits data securely over the communication network.
- **Manage:** Authenticates, monitors, and controls IoT devices.
- **Acquire:** Stores IoT data in databases and big data repositories.
- **Organise and Analyse:** Analyzes stored data to generate insights and support decision-making.

IBM IOT ARCHITECTURE



IBM IOT ARCHITECTURE

Gather + Consolidate + Connect + Collect + Assemble + Manage and Analyse =IoT

Level	Component	Main Function
Level 1	Sensors	Collect real-world data
Level 2	Gateway	Aggregate, filter, and preprocess data
Level 3	Communication Management	Secure data transmission using MQTT and messaging services
Level 4	Device Management	Register, monitor, and manage IoT devices
Level 5	Cloud Services	Store, process, analyze, and visualize IoT data

IBM VS. CISCO IOT ARCHITECTURE

1. Architecture Focus

- **IBM:** Designed for **device connectivity, messaging, cloud analytics, and real-time data processing.**
- **Oracle:** Designed for **enterprise applications**, integrating IoT data with business systems such as ERP, SCM, and CRM.

2. Gateway Layer

- **IBM:** Focuses on **data consolidation** from multiple sensors.
- **Oracle:** Focuses on **data enrichment**, application support, and communication proxies before transmission.

3. Device Management

- **IBM:** Basic device registration and subscription management.
- **Oracle:** Comprehensive device lifecycle management, including: Device identity management, Access management, Device database, Device administration

IBM VS. CISCO IOT ARCHITECTURE

5. Data Storage

- **IBM:** Optimized for **time-series IoT data** using Informix Time Series and NoSQL databases.
- **Oracle:** Combines **Big Data, Oracle RDBMS, and Key-Value Stores** to support both structured and unstructured enterprise data.

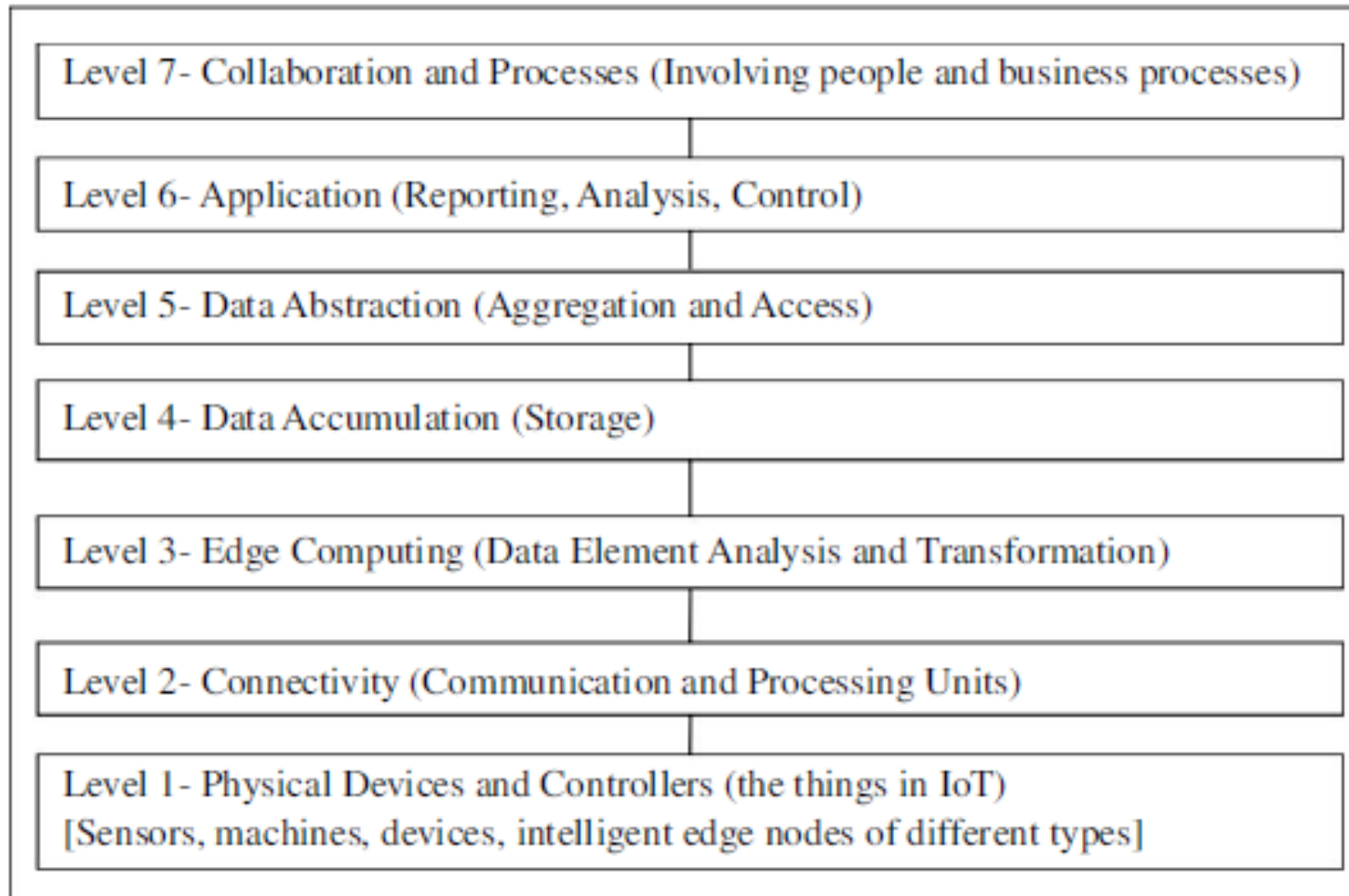
6. Analytics

- **IBM:** Strong emphasis on **real-time analytics**, predictive analytics, and in-memory intelligence.
- **Oracle:** Emphasizes **business intelligence** and enterprise reporting integrated with organizational workflows.

7. Enterprise Integration

- **IBM:** Primarily focuses on IoT platform capabilities.
- **Oracle:** Provides **Enterprise Integration and SOA (Service-Oriented Architecture)** for seamless integration with enterprise software.

CISCO IOT ARCHITECTURAL VIEW



7-LEVEL IOT MODEL USING A FOOD DELIVERY APP

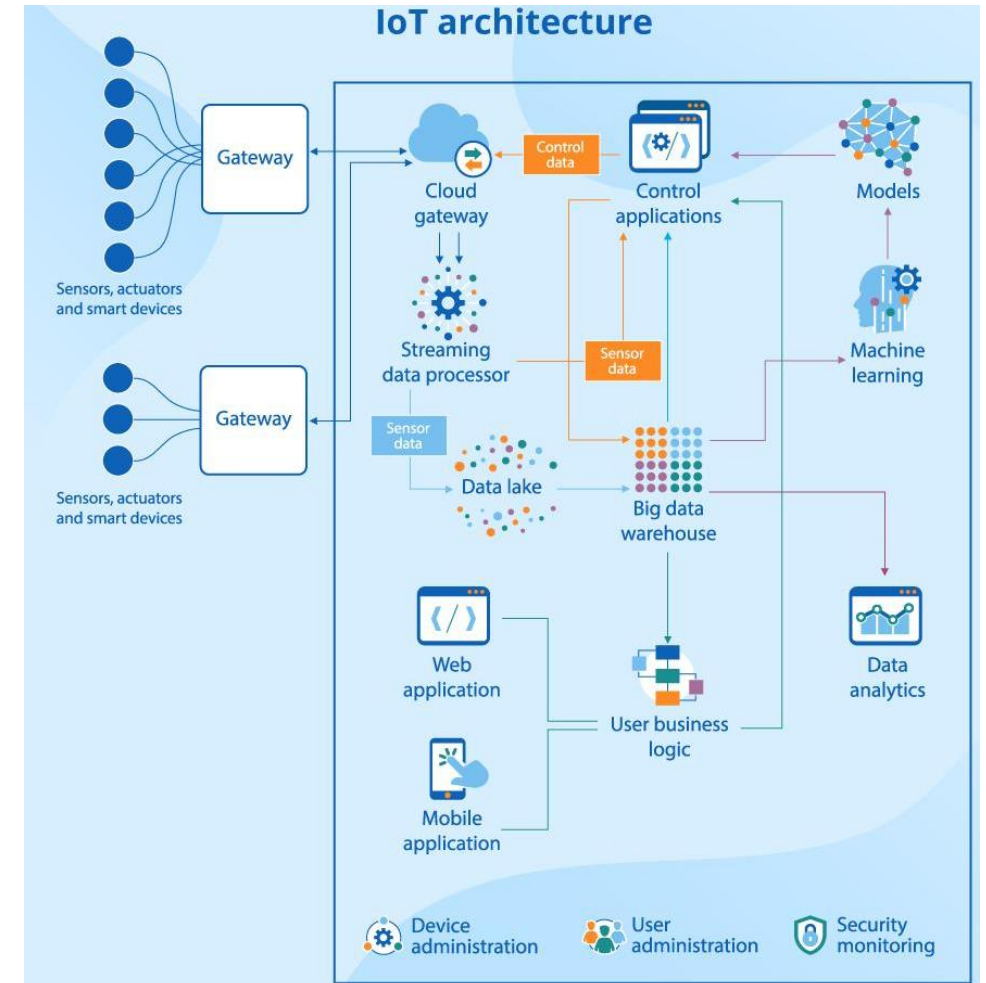
IoT Level	Food Delivery Example
Level 1	Restaurant prepares food (sensor collects data)
Level 2	Delivery partner receives the order (communication network)
Level 3	Delivery app checks traffic and chooses the best route (edge computing)
Level 4	Order details stored in database
Level 5	App combines customer, restaurant, and GPS data
Level 6	Customer tracks the order on the app
Level 7	Company analyzes delivery times and customer satisfaction to improve business

CLASS ROOM ACTIVITY

- Choose any everyday IoT system (smart home, smartwatch, connected car, ATM, smart traffic lights, or fitness band) and identify what happens at each of the seven layers.

COMPONENTS OF IOT SYSTEM

Component	Function
Devices	Sense physical conditions and perform actions
Connectivity	Transfers data between devices and the cloud
Edge/Gateway	Processes data locally and connects devices
Cloud	Stores and manages IoT data
Analytics	Converts data into meaningful insights
Applications	Provides monitoring and control interfaces
Security	Protects devices, data, and communications



COMPONENTS OF IOT SYSTEM

Hardware (Things/Devices)

Purpose: Interface with the physical environment.

Components

- Sensors (temperature, humidity, pressure, motion, GPS, camera)
- Actuators (motors, relays, valves, LEDs)
- Microcontrollers/Microprocessors (Arduino, ESP32, Raspberry Pi, STM32)
- Power supply (battery, solar, power management)

Role: Collect environmental data and execute physical actions.

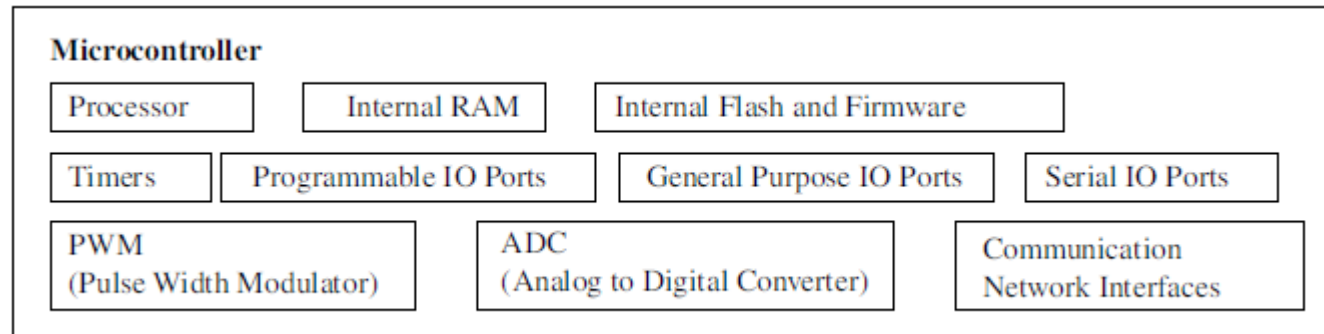


Fig. Various functional units in an MCU that are embedded in an IoT device or a physical object

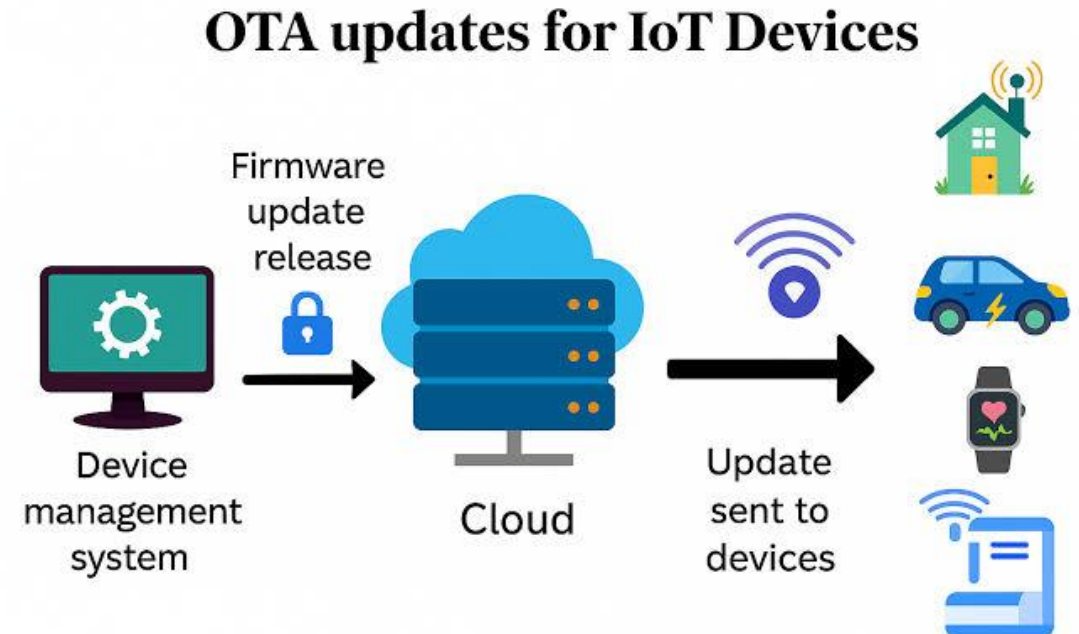
COMPONENTS OF IOT SYSTEM

Firmware

Purpose: Low-level software embedded in IoT devices.

Functions

- Sensor interfacing
- Device initialization
- Data acquisition
- Memory management
- Communication protocol implementation
- Power optimization
- Over-the-air (OTA) firmware updates



COMPONENTS OF IOT SYSTEM

Operating System (OS)

Purpose: It is specialized software that manages hardware resources, memory, communication, and application execution on IoT devices while optimizing power consumption.

Functions

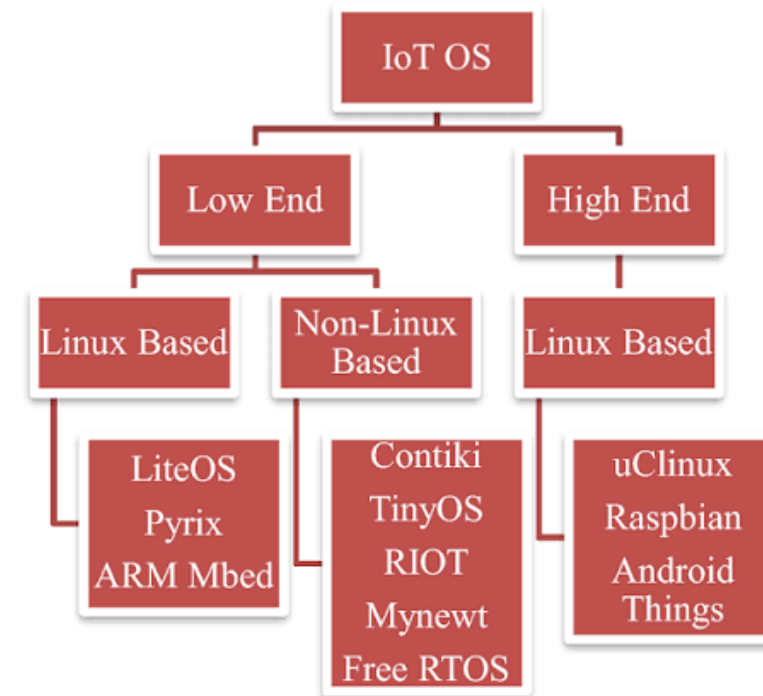
Task scheduling, Memory management, Process management
Networking support, Security mechanisms

1. Low-End IoT Operating Systems

These operating systems are designed for **resource-constrained devices** with limited RAM, flash memory, and CPU power. (Eg: Sensor nodes, wearables)

2. High-End IoT Operating Systems

These operating systems run on **high-performance embedded processors** such as Raspberry Pi, BeagleBone, NVIDIA Jetson, or industrial gateways. (Eg: Smart gateways, edge AI)



COMPONENTS OF IOT SYSTEM

Communication Module

- A **communication module** is the hardware and software component of an IoT device that enables **data transmission and reception** between sensors, actuators, gateways, cloud platforms, and other IoT devices.
- It acts as the **communication interface** of the IoT system by converting sensor data into wireless or wired signals and transmitting them through suitable communication protocols.

Main Functions

- Collects data from sensors or controllers.
- Transmits data to gateways or cloud servers.
- Receives commands from users or cloud applications.
- Supports different communication protocols.
- Ensures reliable and secure data transfer.
- Optimizes energy consumption during communication.

COMPONENTS OF IOT SYSTEM

Types of Communication Technologies

1. Short-Range Communication

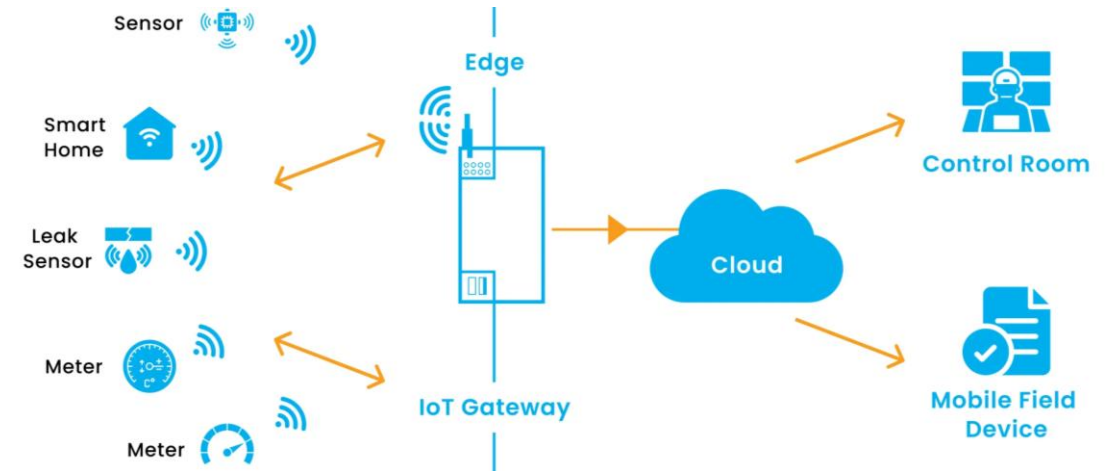
Technology	Range	Speed	Power	Applications
Bluetooth	10–100 m	Medium	Very Low	Wearables, healthcare
BLE	10–100 m	Low	Very Low	Fitness trackers
ZigBee	10–100 m	Low	Low	Home automation
Wi-Fi	50–100 m	High	High	Smart homes, cameras
NFC	<10 cm	Low	Very Low	Payments, access control

COMPONENTS OF IOT SYSTEM

Types of Communication Technologies

2. Long-Range Communication

Technology	Range	Data Rate	Applications
LoRaWAN	2–15 km	Low	Agriculture
Sigfox	10–40 km	Very Low	Environmental monitoring



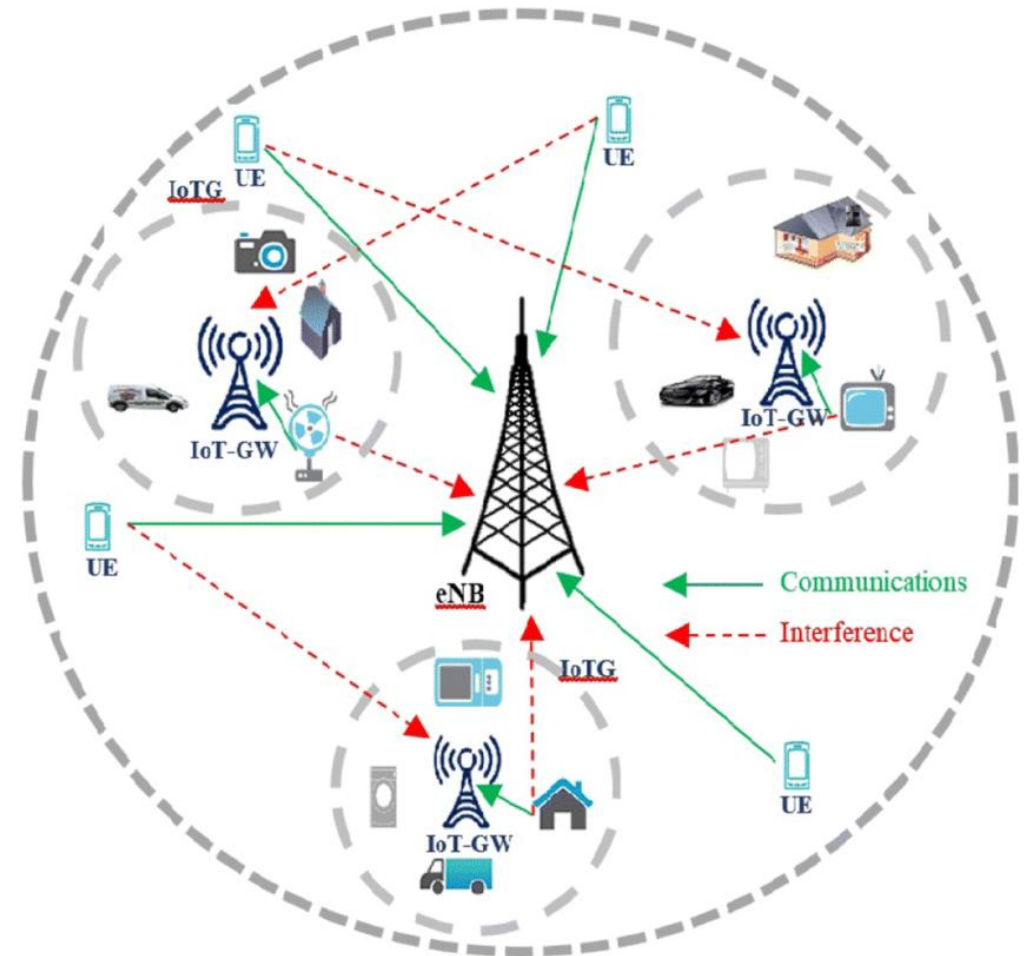
Note: The global 0G Network, powered by Sigfox 0G technology, is a low power wide area network (LPWAN) dedicated to Massive IoT. It is designed to connect devices securely at low cost in the most energy-efficient way.

COMPONENTS OF IOT SYSTEM

Types of Communication Technologies

3. Cellular Communication

Technology	Coverage	Applications
LTE-M	Nationwide	Asset tracking
4G LTE	Wide	Video surveillance
5G	Ultra-wide	Autonomous vehicles



COMPONENTS OF IOT SYSTEM

Types of Communication Technologies

4. Wired Communication

- **Wired communication** is a communication method in which IoT devices exchange data through **physical transmission media**, such as copper cables or optical fiber, instead of wireless signals.
1. **Ethernet** is the most widely used wired communication technology for connecting IoT devices in Local Area Networks (LANs).
 2. **RS-485** is a serial communication standard designed for long-distance industrial communication. (upto 1200m)
 3. **CAN Bus** is a robust communication protocol developed primarily for automotive and industrial applications.
 4. **Modbus** is an industrial communication protocol commonly used for communication between PLCs, sensors, and SCADA systems.
 5. **EtherCAT (Ethernet for Control Automation Technology)** is a real-time industrial Ethernet protocol designed for high-speed automation.

COMPONENTS OF IOT SYSTEM

Types of Communication Technologies

4. Wired Communication

Advantages of Wired Communication

- High reliability and stable communication
- Low latency for real-time applications
- High data transmission speed
- Better security than wireless networks
- Minimal signal interference
- Consistent performance in industrial environments
- Suitable for mission-critical systems

Limitations

- Higher installation and maintenance costs
- Limited mobility due to cables
- More difficult to expand over large areas
- Cable damage can interrupt communication
- Installation can be time-consuming

COMPARISON OF COMMUNICATION TECHNOLOGIES

Wireless Standard	Power	Transmission Range (typical)	Data Rates
Bluetooth	Medium	1 to 100 m	1 to 3 Mbps
Bluetooth LE	Lower	>100 m	125 kbps to 2 Mbps
LoRaWAN	Low	10 km	0.3 to 50 kbps
NB-IoT	Low	<35 km	20 kbps to 5 Mbps
NFC	Low	<10 cm	106 to 424 kbps
Sigfox	Low	3 to 50 km	100 to 600 bps
6LoWPAN	Low	100 m	0 to 250 kbps
802.11/Wi-Fi	Medium	100 m to several km (with boosters)	10 to 100+ Mbps
802.15.4/Zigbee	Low	10 to 100 m	20 to 250 kbps

IOT COMMUNICATION PROTOCOL STACK

Layer	Main Function	Example Protocols
Application	User services and messaging	MQTT, CoAP, HTTP, HTTPS, AMQP, DDS
Transport	End-to-end communication	TCP, UDP, QUIC
Network	Addressing and routing	IPv4, IPv6, 6LoWPAN, RPL
Data Link (MAC)	Framing and medium access	IEEE 802.15.4, Ethernet, Wi-Fi MAC, BLE MAC
Physical	Signal transmission	Wi-Fi, BLE, ZigBee, LoRa, NB-IoT, LTE-M, 5G, Ethernet

PHYSICAL LAYER

The **Physical Layer** is the lowest layer of the communication protocol stack. It is responsible for transmitting and receiving **raw bits** through a physical medium such as radio waves, electrical signals, or optical signals.

Functions: Signal transmission, Signal reception, Frequency selection, Modulation and demodulation, Data encoding, Power control

Communication Technologies:

Technology	Range
Wi-Fi	50–100 m
BLE	10–100 m
ZigBee	10–100 m
LoRa	2–15 km
NB-IoT	Wide area
5G	Wide area

DATA LINK LAYER

The Data Link Layer provides reliable communication between two directly connected devices. It packages raw bits into frames, detects transmission errors, and controls access to the communication medium.

Functions: Frame creation, MAC addressing, Error detection (CRC), Medium access control, Flow control

Communication Technologies: ZigBee, 6LoWPAN, Wi-Fi MAC, BLE MAC

NETWORK LAYER

The Network Layer is responsible for logical addressing and routing of packets between different networks. It determines

- Where the data should go?
- Which route should be followed?

Functions: Packet routing, Packet forwarding, Congestion handling

Communication Technologies: IPv4, IPv6, 6LoWPAN

TRANSPORT LAYER

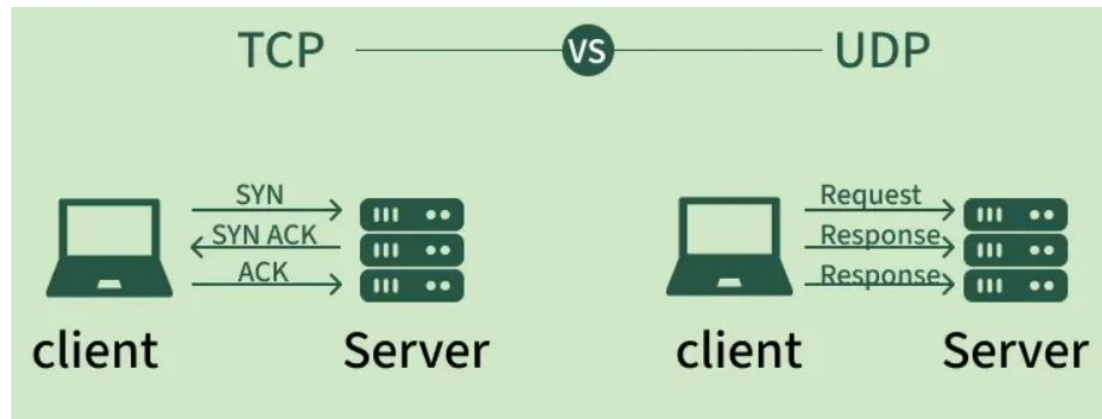
The Transport Layer provides end-to-end communication between source and destination devices. It ensures that the transmitted data reaches the destination correctly.

Functions: Reliable communication, Error recovery, Flow control, Sequencing and Retransmission

Transport layer protocols:

Transmission Control Protocol(TCP): TCP is connection-oriented protocol, so first the connection is established between two ends and then data is transferred and then the connection is terminated after all data being sent.

2. User Datagram Protocol (UDP): When speed and size are more important than security and dependability, UDP is employed.



APPLICATION LAYER

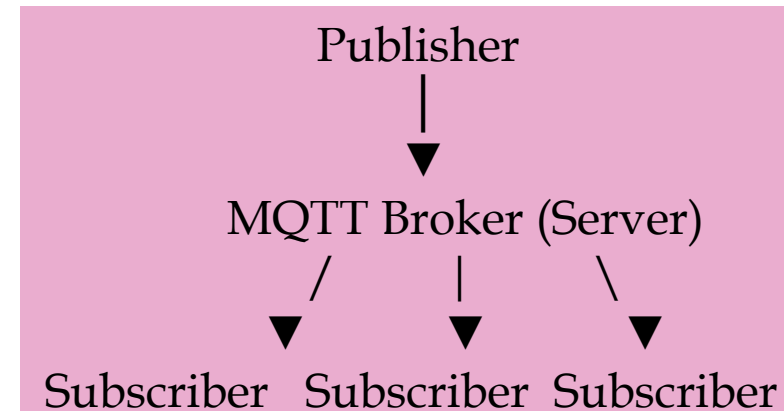
The Application Layer is the highest layer of the protocol stack. It provides services that allow IoT applications, users, and cloud platforms to exchange information. This is the layer where IoT applications interact with the network.

Functions:

- **Data formatting:** Turns raw numbers from sensors into readable messages.
- **Command handling:** Sends user instructions back down to physical actuators or switches.
- **Session control:** Starts and stops secure connections between devices and servers.

Primary protocols: **MQTT (Message Queuing Telemetry Transport):** MQTT is a lightweight, publish-subscribe messaging protocol designed for IoT devices with limited processing power, memory, and bandwidth.

- **Publisher:** Sends messages (e.g., a temperature sensor).
- **Broker:** Receives and distributes messages to subscribers.
- **Subscriber:** Receives messages of interest (e.g., mobile app).



APPLICATION LAYER PROTOCOLS

Primary protocols: MQTT (Message Queuing Telemetry Transport)

Features

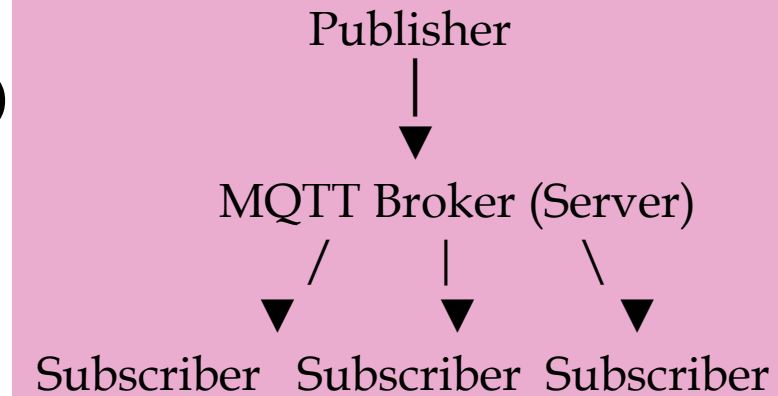
- Publish-Subscribe communication model
- Lightweight protocol with a small packet size
- Uses **TCP**
- Supports **Quality of Service (QoS)** levels

Advantages

- Low power consumption
- Efficient for low-bandwidth networks
- Supports many connected devices
- Reliable message delivery

Limitations

- Requires an MQTT broker
- Less suitable for large multimedia data



APPLICATION LAYER PROTOCOLS

Protocol	Explanation
CoAP (Constrained Application Protocol)	A lightweight request-response protocol designed for low-power and resource-constrained IoT devices. It is commonly used in wireless sensor networks and smart agriculture.
HTTP (Hypertext Transfer Protocol)	A standard client-server protocol used to exchange data between IoT devices and web or cloud servers. It is widely used for web applications
AMQP (Advanced Message Queuing Protocol)	A reliable message queuing protocol that guarantees secure and dependable message delivery. It is commonly used in enterprise IoT and cloud messaging systems.
DDS (Data Distribution Service)	A high-performance publish-subscribe protocol that enables real-time communication between devices without a central broker. It is widely used in robotics, autonomous vehicles, and industrial automation.

IOT APPLICATIONS

Class Activity 1: Think of five IoT devices or services you have used in the last week.

Identify:

- What is the IoT device?
- What data does it collect?
- How does it benefit the user?
- How does it benefit the business?

IOT APPLICATIONS

Class Activity 2:

Business Problem	Suggest an IoT-based Solution
Long queues in supermarkets	
Food wastage in restaurants	
Traffic congestion	
Hospital patient monitoring	
Water wastage in agriculture	
Warehouse inventory errors	